

Notes: Yagan (AER, 2015)

Capital Tax Reform and the Real Economy: The Effects of the 2003 Dividend Tax Cut

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1 Big picture

The paper studies one of the central questions in public finance and corporate finance: **do cuts in shareholder-level capital taxes stimulate real corporate activity?** The specific policy experiment is the 2003 U.S. dividend tax cut, which lowered the top federal dividend tax rate from 38.6 percent to 15 percent. The policy was explicitly promoted as a way to raise investment, hiring, and wages.

The main empirical challenge is that aggregate investment is cyclical. Investment rose after 2003, but the economy was also recovering from the early-2000s recession. Yagan's solution is to compare corporations that were directly affected by the dividend tax cut with corporations that were not directly affected.

The paper's headline finding is stark: **the dividend tax cut caused no detectable increase in corporate investment or employee compensation.** The same empirical design finds an immediate and statistically strong increase in payouts to shareholders, which shows that the tax cut was salient and behaviorally relevant. The null real effect is therefore not because firms ignored the tax cut; it is because the payout response did not translate into more real activity.

2 Incremental contribution of the paper

This paper's contribution is incremental in several dimensions.

1. **Direct test of real effects of a major capital tax reform.** Earlier work on dividend taxes often focused on asset prices, payouts, or structural investment equations. This paper directly tests investment and employee compensation using tax-return microdata.
2. **Quasi-experimental design using C- versus S-corporations.** C-corporations are subject to shareholder dividend taxation, whereas S-corporation dividends are not taxed at the entity/shareholder level in the same way. S-corporations therefore provide a business-cycle control group for C-corporations.
3. **Large administrative tax data.** The analysis uses IRS Statistics of Income corporate tax returns, which include private corporations and are much broader than Compustat.
4. **High-precision null.** The estimates are close to zero and precise enough to challenge the joint hypothesis that the dividend tax cut sharply lowered the cost of capital and that investment is highly responsive to the cost of capital.
5. **Separates financial response from real response.** The paper shows payouts rose sharply, but investment and employee compensation did not. This distinction is the core economic message.
6. **Policy interpretation.** The findings imply that it may be difficult to implement a dividend tax cut with substantially larger near-term real effects, especially if marginal investment is financed by retained earnings and debt rather than newly issued equity.

3 Data and measurement

3.1 Corporate tax data

The paper uses annual U.S. corporate income tax returns from the IRS Statistics of Income sample. The main analysis sample covers years 1998–2008 and includes private C- and S-corporations satisfying the following restrictions:

- assets between \$1 million and \$1 billion;
- revenue between \$0.5 million and \$1.5 billion;
- private through the previous year;
- nonfinancial and nonutility industries;
- valid nonmissing values for the main variables.

The main unbalanced panel contains 333,029 firm-year observations from 73,188 corporations. Approximately 58 percent of the firms are C-corporations. The firm-size range is economically important because firms in this range employ a large share of U.S. private-sector workers.

3.2 Treatment and control groups

The treatment group is C-corporations. These corporations pay corporate income tax, and shareholders pay dividend taxes when earnings are distributed.

The control group is S-corporations. These firms have the same corporate legal form but are flow-through entities for federal tax purposes. Business income passes through to owners' individual returns, and S-corporation dividends are not directly affected by the 2003 dividend tax cut.

The key empirical comparison is not based on random assignment into C or S status. Instead, the identifying assumption is that similar C- and S-corporations would have followed parallel trends absent the dividend tax cut.

3.3 Main variables

- **Investment:** cost of newly purchased tangible capital assets.
- **Net investment:** annual dollar change in tangible capital assets.
- **Employee compensation:** non-officer wages, salaries, benefits, and pension contributions.
- **Payouts:** dividends plus share buybacks.
- **Equity issued:** nonnegative annual changes in paid-in capital.

Outcomes are scaled by lagged tangible capital or lagged revenue, depending on the outcome, and winsorized to reduce the influence of extreme outliers. Observations are dollar-weighted, and the S-corporation sample is reweighted to match the C-corporation distribution across industry-size bins.

4 Models and empirical specifications

4.1 Economic models of dividend taxation

The paper interprets the evidence through two broad classes of dividend tax models.

Traditional view. Dividend taxes raise the cost of capital when firms finance marginal investment by issuing equity or risky claims. A dividend tax cut should lower the cost of capital, increase investment, and potentially raise labor demand.

New view or trapped-equity view. Marginal investment is financed using retained earnings and riskless debt. Earnings will eventually be taxed when distributed whether they are paid out today or retained and paid out later. A dividend tax cut raises both the after-tax return on investment and the opportunity cost of retaining funds, leaving marginal investment unchanged.

The paper’s null investment result is more naturally consistent with the new-view logic, though the paper also discusses ways in which traditional models might explain the null because of policy details such as tax-cut expiration risk.

4.2 Main difference-in-differences specification

The core regression is

$$Y_{it} = \alpha_1 CCorp_{i,t-2} + \alpha_2 (CCorp_{i,t-2} \times Post_t) + X'_{i,t-2} \beta + \gamma_t + \varepsilon_{it}, \quad (1)$$

where Y_{it} is investment, net investment, employee compensation, or payouts. $CCorp_{i,t-2}$ indicates whether the firm was a C-corporation two years before the outcome year. $Post_t$ equals one for 2003 and later. The coefficient of interest is α_2 , the C-corporation change after 2003 relative to S-corporations.

The baseline controls include two-digit NAICS industry fixed effects and quartics in firm age, lagged revenue, lagged profit margin, and revenue growth. Year fixed effects absorb aggregate shocks.

4.3 Elasticity calculation

The paper reports elasticities with respect to one minus the top statutory dividend tax rate:

$$\varepsilon = \frac{\hat{\alpha}_2 / \bar{Y}_C^{pre}}{\Delta(1 - \tau_{div}) / (1 - \tau_{div}^{pre})}. \quad (2)$$

The statutory top dividend tax rate falls from 44.7 percent to 20.8 percent when federal and average state rates are combined. Thus, the policy substantially raises $1 - \tau_{div}$.

4.4 Payout event specification

For payouts, the paper also estimates year-specific C-corporation effects:

$$Payouts_{it} = \alpha_1 CCorp_{i,t-2} + X'_{i,t-2}\beta + \gamma_t + (CCorp_{i,t-2} \times Year_t)' \delta + \varepsilon_{it}. \quad (3)$$

This specification asks whether payouts rise immediately in 2003 and whether the increase persists.

5 Identification and empirical design

The key identifying assumption is **common trends**: absent the dividend tax cut, C- and S-corporations in the same industry-size cells would have followed similar trends.

The paper supports this assumption in several ways:

- C- and S-corporations in the sample overlap substantially by industry and size.
- Main outcomes trend similarly for the two groups before 2003.
- The empirical design reweights S-corporations to match the C-corporation distribution across 190 industry-size bins.
- Accelerated depreciation and other contemporaneous policies applied similarly to both groups or are tested directly as confounds.
- Results are robust in balanced panels with firm fixed effects, where the only changing firm-level variable is the outcome.

A potential concern is that C- and S-corporations differ in levels of profitability, payout policy, or ownership. The design does not require equality in levels; it requires parallel trends after conditioning on controls.

6 Main results

6.1 Investment

The paper estimates essentially zero effect on gross investment. In the main controlled specification, the coefficient is -0.0002 dollars per dollar of lagged tangible capital, with a standard error of 0.0042 . The implied elasticity of investment with respect to $1 - \tau_{div}$ is 0.00 , with a 95 percent confidence interval of approximately $[-0.08, 0.08]$.

6.2 Net investment and employee compensation

The results are also null for net investment and employee compensation. Employee compensation has a negative point estimate and a tight upper bound, implying little evidence that the dividend tax cut raised labor earnings through higher investment or labor demand.

6.3 Payouts

Payouts respond immediately and strongly. C-corporation payouts rise by about 21 percent in 2003 relative to S-corporations in the main event estimate, and the post-2003 payout effect remains large and significant. This demonstrates that the tax cut was salient and relevant for affected firms.

6.4 Heterogeneity and external validity

The paper finds no evidence that the average null masks positive effects among particular firm types. Effects are close to zero across size deciles and across firm subgroups defined by size, age, growth, profitability, cash, and leverage. Adding public corporations in the relevant size range does not materially change the result.

6.5 Economic interpretation

The null real effect rejects the combination of a large traditional-view cost-of-capital effect and a large cost-of-capital elasticity of investment. The result is consistent with new-view/trapped-equity logic or with policy details that prevented the tax cut from reducing marginal investment incentives.

7 Tables and figures

7.1 Figure 1. Industry and size distribution of the main analysis sample and Table 1. Summary statistics for C- and S-corporations

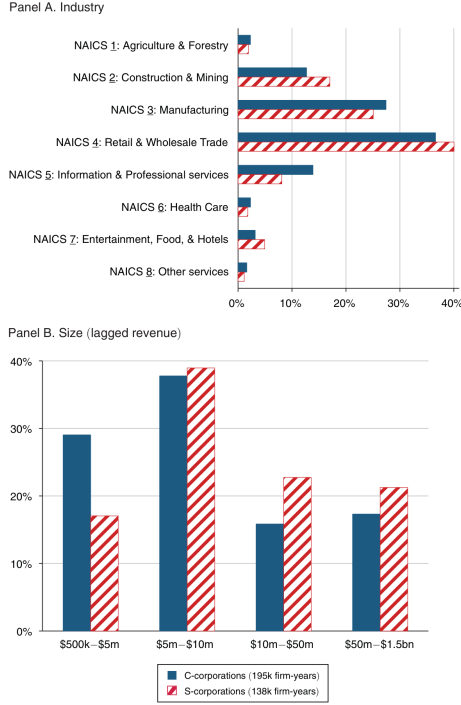


FIGURE 1. INDUSTRY AND SIZE DISTRIBUTION OF THE MAIN ANALYSIS SAMPLE

Notes: This figure plots the industry and size mix of the C-corporations (whose dividends are taxable) and S-corporations (whose dividends are not taxable) in this paper's main analysis sample. Each graph's bars sum to 100 percent within corporation type. Lagged revenue denotes operating revenue averaged over the preceding two lags. This sample is an unbalanced panel of annual corporate income tax returns, comprising all observations from the IRS Statistics of Income stratified random sample in years 1998–2008 in which the filing corporation had between \$1 million and \$1 billion in lagged assets and \$500,000 and \$1.5 billion in lagged revenue, was private through the previous year, and is not in the finance or utilities industries. All analyses flexibly control for any time-varying industry or firm-size shocks by non-parametrically reweighting the S-corporation sample within every year to match the distribution of C-corporations across 190 industry-firm-size bins as detailed in Section III E. C- versus S-status is defined as of the second lag; corporations can switch status if they meet the legal requirements but

Figure 1. Industry and size distribution of the main analysis sample.

TABLE 1—UNWEIGHTED SUMMARY STATISTICS FOR THE MAIN ANALYSIS SAMPLE

	C-corporations				S-corporations			
	Mean (1)	Median (2)	10th percentile (3)	90th percentile (4)	Mean (5)	Median (6)	10th percentile (7)	90th percentile (8)
<i>Characteristics</i>								
Lagged revenue	69,214,032	26,410,150	3,310,941	164,050,464	76,377,272	42,265,004	5,385,821	169,980,800
Lagged assets	45,330,360	16,945,392	1,878,245	105,045,088	35,529,524	19,258,636	3,002,156	74,923,672
Lagged tangible capital assets	10,803,074	2,041,562	118,378	25,007,676	7,826,240	2,281,994	173,325	17,449,492
Lagged profit margin	−0.03	0.04	−0.09	0.17	0.08	0.06	−0.01	0.25
Lagged revenue growth	0.15	0.03	−0.21	0.45	0.10	0.03	−0.18	0.34
Lagged cash/lagged assets	0.18	0.10	0.01	0.47	0.17	0.10	0.01	0.46
Lagged leverage	0.68	0.66	0.21	1.00	0.63	0.66	0.16	0.97
Age	26	22	6	52	27	23	7	51
<i>Outcomes</i>								
Investment	2,245,204	249,801	1,185	4,599,334	1,909,465	308,066	4,502	3,803,072
Investment/lagged tangible capital assets	1.608	0.153	0.001	0.767	1.112	0.166	0.005	0.791
Net investment	440,842	−19,591	−1,280,729	1,756,233	349,969	−21,495	−988,169	1,572,145
Net investment/lagged tangible capital assets	0.870	−0.034	−0.286	0.459	1.715	−0.030	−0.254	0.454
Employee compensation	12,410,943	3,843,863	324,038	28,162,686	11,265,016	5,013,783	452,621	24,175,638
Employee compensation/lagged revenue	0.291	0.160	0.028	0.492	0.188	0.131	0.027	0.376
Payouts	659,858	0	0	443,330	3,486,271	684,450	0	7,762,084
Payouts/lagged revenue	0.015	0.000	0.000	0.011	0.093	0.016	0.000	0.169
Dividends	531,236	0	0	250,080	3,410,537	658,080	0	7,599,040
Dividends/lagged revenue	0.012	0.000	0.000	0.006	0.092	0.016	0.000	0.166
Equity issued	2,754,047	0	0	572,898	276,790	0	0	389
Equity issued/lagged revenue	0.239	0.000	0.000	0.013	0.023	0.000	0.000	0.000
Number of firm-year observations	195,033				137,996			
Number of firms	43,988				32,113			

Notes: This table lists unweighted summary statistics for C-corporations (whose dividends are taxable) and S-corporations (whose dividends are not taxable) in this paper's main analysis sample: an unbalanced panel of annual corporate income tax returns, comprising all observations from the IRS Statistics of Income stratified random sample in years 1998–2008 in which the filing corporation had between \$1 million and \$1 billion in lagged assets and \$500,000 and \$1.5 billion in lagged revenue, was private through the previous year, and is not in the finance or utilities industries. Lagged denotes averaged over the two preceding lags. Revenue equals operating revenue. Assets equals the book value of assets. Tangible capital assets, also called capital, equals the book value of tangible capital assets (e.g., excluding cash and patents). Profit margin equals one minus the ratio of operating costs to revenue. Cash equals liquid current assets. Leverage equals the book value of non-equity liabilities divided by assets (this is greater than one when accumulated losses exceed paid-in equity). Age equals the year of the return minus the year of incorporation. Investment equals the cost of all newly purchased tangible capital assets. Net investment equals the annual dollar change in tangible capital assets. Employee compensation equals the sum of all non-officer wages, salaries, benefits, and pension contributions. Dividends equals cash plus property distributions to shareholders. Payouts, also called total payouts to shareholders, equals dividends plus share buybacks (non-negative annual changes in treasury stock). Equity issued equals non-negative annual changes in paid-in capital. C- versus S-status is defined as of the second lag; corporations can switch status if they meet the legal requirements but fewer than 4

Table 1. Summary statistics for C- and S-corporations.

Figure 1. Read:

- C- and S-corporations are both present across the same broad industries.
- Both groups also overlap by size, though C-corporations are more common in the smallest revenue bin and S-corporations are somewhat more common in the larger private-firm bins.
- The empirical strategy further reweights S-corporations to match the C-corporation distribution across fine industry-size bins.

Figure 1. Economic message: The treated and control firms are not identical, but they are economically comparable enough to support a common-trends design after flexible industry-size reweighting. The figure makes the core control-group logic visually credible.

Table 1. Read:

- The sample contains 195,033 C-corporation firm-years and 137,996 S-corporation firm-years.
- C- and S-corporations have similar order-of-magnitude revenue, assets, investment, and employee compensation.
- S-corporations have much higher dividends and payouts relative to revenue before the tax cut, which motivates the percentage-style payout scaling later in the paper.

Table 1. Economic message: The sample is composed of economically meaningful private firms, not tiny shell companies. The table also clarifies why outcome scaling and controls are important: levels differ, but the identification relies on changes over time.

7.2 Figure 2. Time-series effects of the 2003 dividend tax cut and Figure 3. Effects by firm-size decile

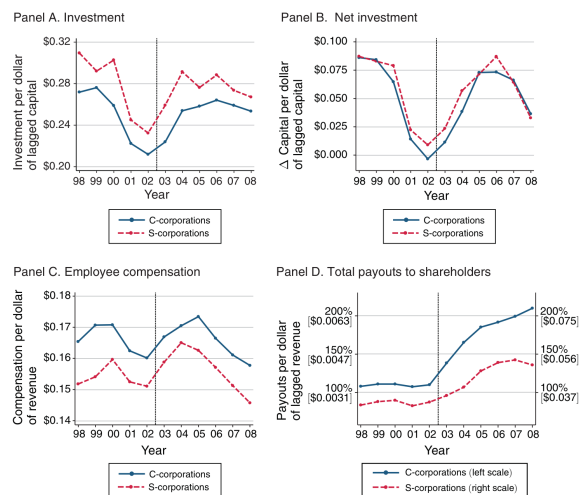


FIGURE 2. EFFECTS OF THE 2003 DIVIDEND TAX CUT

Notes: These figures plot the time series of annual mean outcomes for C-corporations and S-corporations in the main analysis sample net of a rich set of controls. Investment equals the cost of all newly purchased tangible capital assets. Net investment equals the annual dollar change in tangible capital assets. Employee compensation equals the sum of all non-officer wages, salaries, benefits, and pension contributions. Total payouts to shareholders equals dividends plus share buybacks (non-negative annual changes in treasury stock). Each panel is constructed by scaling each observation by either the firm's tangible capital assets or revenue averaged over the two preceding lags; winsorizing (top-coding) observations at the ninety-fifth percentile; regressing this scaled outcome variable within every year on a C-corporation indicator, two-digit NAICS industry fixed effects, and quartics in age, lagged revenue, lagged profit margin, and revenue growth; and requiring that the vertical distance between the two lines equals the regression coefficient on the C-corporation indicator and that the weighted average of the lines equals the sample average in that year. The regressions are dollar-weighted (each observation is weighted by its lagged revenue) and flexibly control for any time-varying industry or firm-size shocks by non-parametrically reweighting the S-corporation sample within every year to match the distribution of C-corporations across 190 industry-firm-size bins as detailed in Section IIE. Panel D is included as a test for an immediate behavioral response in financial outcomes and differs from the other graphs in two ways that account for income-tax-induced differences in baseline payout levels and for slightly differential pre-trends as detailed in Section IVA.

Figure 2. Time-series effects of the 2003 dividend tax cut.

Figure 2. Read:

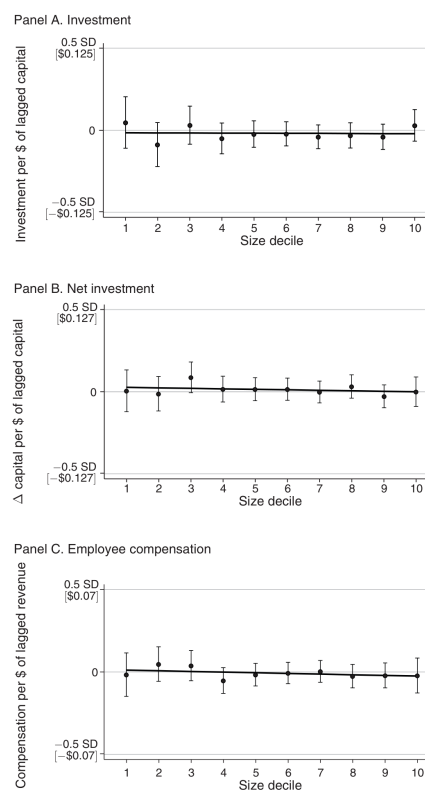


FIGURE 3. EFFECTS BY SIZE DECILE

otes: This figure plots estimated within-size-decile effects of the 2003 dividend tax cut in the main analysis sample. Variables are defined, scaled, and winsorized as detailed in Figure 2. Each y-axis height equals one standard deviation of the outcome. Each panel is computed by binning corporations into deciles according to the unweighted scales of the pooled C-corporation lagged revenue distribution, and then within each decile estimating a regression of the outcome on a C-corporation indicator, the interaction of a C-corporation indicator and post-2003 indicator, year fixed effects, two-digit NAICS industry fixed effects, and quartics in age, lagged revenue, lagged profit margin, and revenue growth. Each panel plots the coefficients on the interaction term with Bonferroni-corrected 95 percent confidence intervals to adjust for multiple (ten) hypothesis testing; uncorrected confidence intervals are one-third

Figure 3. Effects by firm-size decile.

- Panels A–C show investment, net investment, and employee compensation for C- and S-corporations tracking each other closely before and after 2003.
- Panel D shows a sharp relative increase in C-corporation payouts immediately after 2003.
- The real outcomes are flat relative to the control group, while the financial payout outcome moves strongly.

Figure 2. Economic message: This is the central picture of the paper. The policy changed payout behavior but not real activity. The event is salient, but it does not generate the intended investment or labor response.

Figure 3. Read:

- Estimated effects on investment, net investment, and employee compensation are close to zero in every size decile.
- The fitted line across deciles is essentially flat.
- Confidence intervals do not suggest a hidden positive effect concentrated among the largest or smallest firms inside the sample.

Figure 3. Economic message: The null result is not simply an average of offsetting size-specific responses. It appears broadly stable across the firm-size distribution observed in the data.

7.3 Table 2. Difference-in-differences estimates for investment, net investment, and employee compensation and Table 3. Heterogeneity of treatment effects

TABLE 2—EFFECT OF THE 2003 DIVIDEND TAX CUT ON INVESTMENT, NET INVESTMENT, AND EMPLOYEE COMPENSATION

Dependent variable:		Investment					
Dep. var. winsorized at:		95th percentile			99th percentile		
Panel:		Unbalanced	Balanced	Unbalanced	Balanced		
		(\$ per lagged capital)	(\$ per 96-97 cap.)	(\$ per lagged capital)	(\$ per 96-97 cap.)		
		(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Investment</i>							
C-Corp × Post-2003		0.0008 (0.0044)	−0.0002 (0.0042)	−0.0063 (0.0226)	−0.0104 (0.0068)	−0.0118 (0.0066)	−0.1884 (0.1483)
Lagged controls		X			X		
Firm FE's		X			X		
Observations (firm-years)		333,029	333,029	85,624	333,029	333,029	85,624
Clusters (firms)		73,188	73,188	7,784	73,188	73,188	7,784
R ²		0.01	0.07	0.53	0.01	0.05	0.55
Pre-2003 C-corp mean		0.2428	0.2428	0.2939	0.2828	0.2828	0.3682
Pre-2003 C-corp SD		0.2514	0.2514	0.3070	0.4181	0.4181	0.6478
Implied ε wrt $(1 - \tau_{div})$		0.01	0.00	−0.05	−0.09	−0.10	−1.18
		[−0.08, 0.09]	[−0.08, 0.08]	[−0.4, 0.3]	[−0.19, 0.02]	[−0.2, 0.01]	[−3.01, 0.64]
<i>Panel B. Net investment and employee compensation</i>							
Dependent variable:		Net investment			Employee compensation		
Dep. var. winsorized at:		95th percentile			99th percentile		
Panel:		Unbalanced	Balanced	Unbalanced	Balanced		
		(\$ per lagged capital)	(\$ per 96-97 cap.)	(\$ per lagged revenue)	(\$ per 96-97 rev.)		
		(7)	(8)	(9)	(10)	(11)	(12)
C-Corp × Post-2003		0.0048 (0.0041)	0.0042 (0.0039)	−0.0110 (0.0116)	−0.0013 (0.0025)	−0.0013 (0.0020)	0.0083 (0.0062)
Lagged controls		X			X		
Firm FE's		X			X		
Observations (firm-years)		333,029	333,029	85,624	333,029	333,029	85,624
Clusters (firms)		73,188	73,188	7,784	73,188	73,188	7,784
R ²		0.01	0.04	0.20	0.00	0.37	0.87
Pre-2003 C-corp mean		0.0421	0.0421	0.0885	0.1647	0.1647	0.1727
Pre-2003 C-corp SD		0.2541	0.2541	0.2732	0.1415	0.1415	0.1450
Implied ε wrt $(1 - \tau_{div})$		0.26	0.23	−0.29	−0.02	−0.02	0.11
		[−0.18, 0.71]	[−0.19, 0.66]	[−0.88, 0.3]	[−0.09, 0.05]	[−0.07, 0.04]	[−0.05, 0.27]

Notes: This table reports difference-in-differences estimates of the effect of the 2003 dividend tax cut on real outcomes. All columns display the coefficient on the interaction between a C-corporation indicator and an indicator for the year being 2003 or later, from a regression of the outcome on this interaction, a C-corporation indicator, year fixed effects and possibly additional controls. Lagged controls indicates that the regression includes two-digit NAICS industry fixed effects and quartics in age, lagged revenue, lagged profit margin, and revenue growth. Firm FE's indicates that the regression includes firm fixed effects. The unbalanced panel is this paper's main sample; see Table 1 for details. The balanced panel is constructed similarly, except the sample restrictions apply only to years 1996–1997 and observations are required in all years 1998–2008. Before the regression, each observation's outcome value is scaled by either the firm's tangible capital assets or its revenue (see online Appendix Table 2 for alternative scalings) averaged over the two preceding lags in the unbalanced panel and over 1996–1997 in the balanced panel, and then winsorized (top-coded) at the level indicated. The regressions are dollar-weighted (each observation is weighted by its lagged or 1996–1997 revenue) and they flexibly control for any time-varying industry or firm-size shocks by non-parametrically reweighting the S-corporation sample within every year to match the distribution of C-corporations across 190 industry-firm-size bins as detailed in Section IIE. Elasticity equals the reported coefficient divided by the pre-2003 C-corporation outcome mean, divided by the percent change in one-minus-the-top-statutory-dividend-tax-rate (the top rate fell from 44.7 percent to 20.8 percent). Standard errors are clustered by

Table 2. Difference-in-differences estimates for investment, net investment, and employee compensation.

Table 2. Read:

- The main controlled investment estimate is -0.0002 per dollar of lagged capital, essentially zero.
- Net investment estimates are small and statistically insignificant.
- Employee compensation estimates are also statistically insignificant, with tight confidence intervals in the unbalanced panel.
- Results are robust to balanced panels, firm fixed effects, and 95th versus 99th percentile winsorization.

Table 2. Economic message: This table is the paper's main statistical evidence. The estimates are not merely insignificant; they are economically tiny relative to what traditional cost-of-capital models would predict.

TABLE 3—EFFECT HETEROGENEITY

	Investment	Net investment	Employee comp.	Payouts	Equity issued
	(\$ per lagged capital)	(\$ per lagged capital)	(\$ per lagged revenue)	(Percent)	(\$ per lagged revenue)
	(1)	(2)	(3)	(4)	(5)
C-Corp × Post-2003					
× High lagged revenue	0.0103 (0.0127)	−0.0017 (0.0102)	−0.0042 (0.0054)	−3.6 (8.9)	−0.0009 (0.0004)
× High age	0.0104 (0.0168)	0.0003 (0.0144)	−0.0055 (0.0060)	40.0 (10.4)	0.0003 (0.0006)
× High lagged rev. growth	−0.0069 (0.0160)	−0.0164 (0.0165)	−0.0006 (0.0082)	−8.5 (11.0)	−0.0005 (0.0008)
× High profit margin	−0.0265 (0.0167)	0.0103 (0.0140)	−0.0106 (0.0109)	97.9 (16.0)	0.0020 (0.0012)
× High cash/assets	−0.0212 (0.0155)	−0.0217 (0.0148)	−0.0120 (0.0115)	34.7 (12.2)	−0.0006 (0.0011)
× High leverage	−0.0030 (0.0199)	0.0144 (0.0190)	−0.0120 (0.0101)	−59.6 (17.8)	−0.0002 (0.0012)

Notes: This table reports triple-difference estimates of the effect of the 2003 dividend tax cut. Each cell represents a separate regression and reports the coefficient on the triple interaction of a C-corporation indicator, an indicator for the year being 2003 or later, and an indicator for the firm being in the top quintile rather than the bottom quintile (the middle three quintiles are omitted) of the trait specified in the row heading (see Table 1 for definitions). The specifications underlying each cell of columns 1–3 are identical to the difference-in-differences specifications underlying Table 2 columns 2, 8, and 11, respectively, except that each regression fully interacts the top-quintile indicator with the C-corporation and post-2003 indicators. Similar to Table 2, regressions are dollar-weighted (each observation is weighted by its lagged revenue) and flexibly control for any time-varying industry and firm-size shocks by non-parametrically reweighting the S-corporation sample within every year and quintile to match the distribution of C-corporations across 190 industry-firm-size bins; the exception is regressions by the lagged-revenue trait which can be reweighted only across 19 industry bins since the top and bottom quintiles do not overlap in size. Column 4 makes the same modifications to the difference-in-difference regression underlying Table 4 column 2. Column 5 replicates this table's column 3 for the outcome of equity issued. Standard errors are clustered by firm.

Table 3. Heterogeneity of treatment effects.

Table 3. Read:

- Triple-difference estimates by high revenue, age, growth, profit margin, cash, and leverage show no consistent positive investment or compensation response.
- Payout effects vary across firm types but do not line up with offsetting equity issuance patterns.
- The table does not support an efficiency-reallocation story in which some firms cut wasteful investment while others raise productive investment.

Table 3. Economic message: The null real effect is not hiding in firm heterogeneity. If anything, the heterogeneity evidence strengthens the interpretation that the dividend tax cut mainly affected financial distributions rather than real investment allocation.

7.4 Table 4. Effect of the 2003 dividend tax cut on total payouts to shareholders

TABLE 4—EFFECT OF THE 2003 DIVIDEND TAX CUT ON TOTAL PAYOUTS TO SHAREHOLDERS (*Percent*)

Panel:	Unbalanced		Balanced	Unbalanced		Balanced
	(1)	(2)	(3)	(4)	(5)	(6)
C-Corp $\times$ Post-2003	23.4 (3.6)	27.6 (3.3)	78.1 (8.0)	39.4 (7.3)	45.5 (6.5)	53.6 (15.1)
C-Corp $\times$ Year-2003	18.1 (4.3)	21.4 (4.1)	58.5 (8.8)	26.2 (4.8)	30.5 (4.6)	45.1 (11.3)
C-Corp $\times$ Year-2004	32.1 (5.2)	35.6 (5.0)	66.6 (11.4)	43.3 (6.5)	48.3 (6.2)	48.8 (10.4)
C-Corp $\times$ Year-2005	26.8 (5.8)	29.8 (5.5)	81.4 (12.4)	41.2 (8.2)	46.0 (7.5)	59.1 (16.6)
Lagged controls		X			X	
Firm FE's			X			X
Pre-trend controls				X	X	X
Observations (firm-years)	333,029	333,029	85,624	333,029	333,029	85,624
Clusters (firms)	73,188	73,188	7,784	73,188	73,188	7,784
Pre-2003 C-corp mean (\$ per lagged revenue)	0.0031	0.0031	0.0061	0.0031	0.0031	0.0061

Notes: This table reports difference-in-differences estimates of the effect of the 2003 dividend tax cut on total payouts to shareholders (dividends plus buybacks). The first row of columns 1–3 use the same specifications, controls, scaling, weights underlying Table 2 columns 10–12 except that before the winsorizing and in order to account for large level differences in pre-2003 payouts (see Table 1 and the y-axis of Figure 2, panel D), each firm i 's payouts in year t is divided by the mean level of payouts for i 's corporate type (C or S) in the pre-2003 period, essentially transforming the comparison into percentage terms. The second through fourth rows of each column report results from a separate regression in which the C-corp $\times$ post-2003 interaction term is replaced with a vector of interactions between the C-corporation indicator and post-2003 year indicators; see online Appendix Table 8 for additional reported coefficients. Columns 4–6 modify the specifications of columns 1–3 in order to allow for differential pre-2003 trends; see Section IVA for the specification and online Appendix Table 1 for analogous specifications for real outcomes. Standard errors are clustered by firm. See online Appendix 9 for results on the outcome of dividends only.

Table 4. Effect of the 2003 dividend tax cut on total payouts to shareholders.

Table 4. Read:

- C-corporation payouts rise sharply after 2003 relative to S-corporation payouts.
- The immediate 2003 effect is around 21 percent in the main controlled unbalanced-panel estimate.
- The post-2003 average effect is even larger, and results remain strong after controlling for pre-trends.

Table 4. Economic message: The payout response is the paper’s salience check. Firms did respond to the tax cut, but they responded by distributing more cash to shareholders rather than by raising investment or employee compensation.

8 Short summary

- **Question:** Did the 2003 U.S. dividend tax cut stimulate real corporate activity?
- **Design:** Compare C-corporations affected by dividend taxation with S-corporations not directly affected, before and after 2003.
- **Data:** IRS corporate tax returns, 1996–2008, with main analysis on large private firms from 1998–2008.
- **Main real result:** No detectable effect on gross investment, net investment, or employee compensation.
- **Main financial result:** Payouts to shareholders rose immediately and persistently.
- **Interpretation:** The tax cut was salient but did not reduce the relevant marginal cost of capital enough to stimulate near-term real investment, or investment was much less responsive than traditional estimates suggest.

9 Conclusion

Yagan’s paper provides a clean and influential test of whether dividend tax cuts translate into real economic activity. The answer in this setting is no. Using C-corporations as the treated group and S-corporations as a control group, the paper finds that investment and employee compensation did not rise after the 2003 dividend tax cut. The estimated effects are close to zero and precise enough to challenge standard views in which dividend taxes are a powerful near-term investment lever.

The most important nuance is that the tax cut did matter. Payouts increased sharply, so the absence of investment effects cannot be dismissed as a failure of salience or implementation. The policy changed financial behavior, not real behavior.

The incremental contribution is therefore not just a null result. It is a disciplined separation of financial and real corporate responses to a major tax reform. The paper shows that an observed increase in payouts is not sufficient evidence that a capital tax cut stimulated production, employment, or capital formation.

The broader policy implication is cautious. Dividend tax cuts may increase after-tax shareholder income and change distribution policy, but they may not be reliable tools for near-term investment stimulus. The evidence is especially consistent with models in which marginal investment is funded from retained earnings and riskless debt, so shareholder dividend taxes do not determine the marginal investment decision.